

Failure (accidentally blocking the event loop) — Solution

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Basic Version

Approach 1 — the direct way

```
import asyncio
import time

async def good_task():
    await asyncio.sleep(3)
    return "good"

async def blocking_task():
    time.sleep(3)
    return "blocking"

async def time_it(label, coro_a, coro_b):
    start = time.perf_counter()
    results = await asyncio.gather(coro_a, coro_b)
    elapsed = time.perf_counter() - start
    print(label, elapsed, results)

async def main():
    await time_it("two good tasks:", good_task(), good_task())
    await time_it("good + blocking:", good_task(), blocking_task())

asyncio.run(main())
```

Expected output (your exact numbers will vary slightly):

```
two good tasks: 3.01 ['good', 'good']
good + blocking: 6.02 ['good', 'blocking']
```

"Two good tasks" finishes in ~3 seconds — they genuinely overlapped. "Good + blocking" finishes in ~6 seconds — `time.sleep(3)` inside `blocking_task` froze the entire event loop for those 3 seconds, so `good_task` couldn't make any progress either, even though it was written correctly. No error, no warning — the only symptom is the doubled time.

This version works correctly and demonstrates the bug clearly. It's missing type hints and an `if __name__ == "__main__":` guard — both fine for a first working version.

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Intermediate Version

Approach 1 — typed, with a `__main__` guard

```
import asyncio
import time

async def good_task() -> str:
    await asyncio.sleep(3)
    return "good"

async def blocking_task() -> str:
    time.sleep(3)          # the bug: no await, freezes the whole loop
    return "blocking"

async def time_it(label: str, coro_a, coro_b) -> None:
    start = time.perf_counter()
    results = await asyncio.gather(coro_a, coro_b)
    elapsed = time.perf_counter() - start
    print(f"{label} took {elapsed:.2f}s -> {results}")

async def main() -> None:
    await time_it("two good tasks:", good_task(), good_task())
    await time_it("good + blocking:", good_task(), blocking_task())

if __name__ == "__main__":
    asyncio.run(main())
```

Expected output:

```
two good tasks: took 3.01s -> ['good', 'good']
good + blocking: took 6.02s -> ['good', 'blocking']
```

Difference from Basic: full type hints document each function's contract. The `if __name__ == "__main__":` guard means this file can be imported (say, to reuse `time_it` in a later test) without the two 3-6 second demonstration runs firing automatically on import. Same two numbers either way — proving the exact same bug.

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Advanced Version

Approach 1 — `asyncio.to_thread`, the actual fix when you can't avoid a blocking call

```
import asyncio
import time

async def good_task() -> str:
    await asyncio.sleep(3)
    return "good"

async def blocking_task() -> str:
    time.sleep(3)
    return "blocking"

async def fixed_task() -> str:
    await asyncio.to_thread(time.sleep, 3)
    return "fixed"

async def time_it(label: str, coro_a, coro_b) -> None:
    start = time.perf_counter()
    results = await asyncio.gather(coro_a, coro_b)
    elapsed = time.perf_counter() - start
    print(f"{label} took {elapsed:.2f}s -> {results}")

async def main() -> None:
    await time_it("two good tasks:", good_task(), good_task())
    await time_it("good + blocking:", good_task(), blocking_task())
    await time_it("good + fixed (to_thread):", good_task(), fixed_task())

if __name__ == "__main__":
    asyncio.run(main())
```

Expected output:

```
two good tasks: took 3.01s -> ['good', 'good']
good + blocking: took 6.02s -> ['good', 'blocking']
good + fixed (to_thread): took 3.02s -> ['good', 'fixed']
```

`fixed_task` still calls the exact same blocking `time.sleep(3)` internally — nothing about `time.sleep` itself changed. What changed is *where* it runs: `asyncio.to_thread(time.sleep, 3)` hands that call off to a separate worker thread and gives the event loop back something it can genuinely `await`, so `good_task` gets to run concurrently again, same as if `fixed_task` had used `asyncio.sleep` natively.

Approach 2 — the realistic version: a sync-only library call, not `time.sleep`

```
import asyncio
import time
import requests  # a real, sync-only library -- no async version exists

def fetch_sync(url: str) -> int:
    """A normal, blocking function from a library that was never written to be as
    response = requests.get(url, timeout=10)
    return response.status_code

async def good_task() -> str:
    await asyncio.sleep(1)
    return "good"

async def fetch_status_blocking(url: str) -> int:
    # the bug, in its realistic form: calling a sync-only library directly
    # from inside async def, with no await protecting the event loop
    return fetch_sync(url)

async def fetch_status_fixed(url: str) -> int:
    # the fix: the exact same sync function, run off-thread
    return await asyncio.to_thread(fetch_sync, url)

async def main() -> None:
    start = time.perf_counter()
    results = await asyncio.gather(good_task(), fetch_status_fixed("https://example.com"))
    print(f"good + fixed fetch: {time.perf_counter() - start:.2f}s -> {results}")
```

```
if __name__ == "__main__":
    asyncio.run(main())
```

Expected output (timing depends on the real network call, but `good_task`'s 1-second wait genuinely overlaps with the request instead of adding to it):

```
good + fixed fetch: 0.34s -> ['good', 200]
```

This is the realistic version of the bug: nobody writes `time.sleep(3)` inside `async def` in production, but plenty of code accidentally calls a sync-only library (`requests` , a database driver without an async variant, a file-parsing library) the same way `fetch_status_blocking` does here — and it blocks the event loop exactly as badly, just without a name that gives it away.

`fetch_status_fixed` fixes it the same way Approach 1 did: `asyncio.to_thread(...)` around the call, nothing else about the library or the call itself changes.

Difference from Intermediate, and between these 2 Advanced approaches: Intermediate only diagnoses and measures the bug. Approach 1 fixes the exact same `time.sleep(3)` call from the exercise, to prove the mechanism works. Approach 2 shows the fix applied to what this bug actually looks like in real code — a synchronous library function (`requests.get` , standing in for any sync-only dependency) called from inside `async def` without protection — which is the version of this mistake you'll actually make one day, not a `time.sleep()` typo.

Which one should you actually write? In real code: never call a known-blocking function directly from inside `async def` — either use that library's async equivalent if one exists (`httpx` / `aiohttp` instead of `requests` , as Doc09's Core Concepts note), or wrap the sync call in `asyncio.to_thread(...)` the way Approach 2 does when no async version exists. Reach for `asyncio.to_thread` specifically when you're stuck with a sync-only dependency you can't swap out — it's the standard, minimal-effort fix, and exactly what protects an agent's event loop the moment one of its tools turns out to be a synchronous library call in disguise.